FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

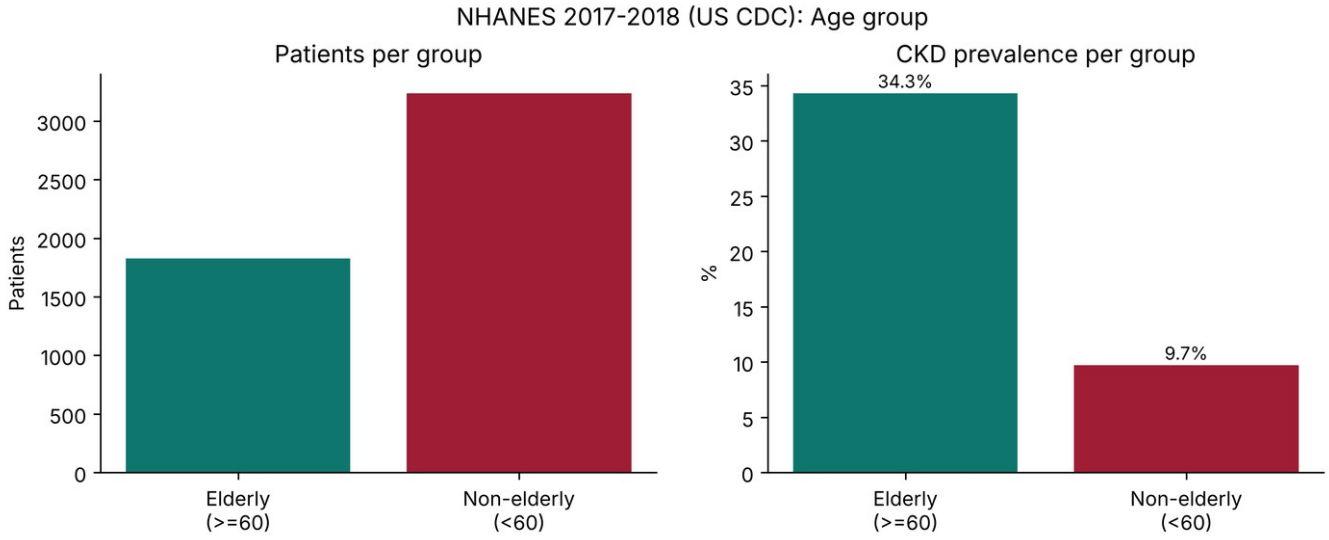
	Value
Dataset	NHANES 2017-2018 (US CDC)
Feature set	Routine blood tests (17 features, no kidney-function tests)
Sensitive attribute	Age group
Patients	5063
Features	17
Groups	Elderly (>=60), Non-elderly (<60)
Decision threshold	training-set CKD prevalence
CV protocol	stratified 5-fold, 5 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **Random Forest** (accuracy 0.740, balanced accuracy 0.737, AUC 0.806).
• Bias detection: in **7 of 7** models, CKD is detected least often in the **Non-elderly (<60)** group (Equal Opportunity gap 0.374–0.561).
• Even a perfect classifier has DP diff = 0.246 and DI = 0.282 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.539 → 0.471, balanced accuracy 0.736 → 0.732.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Unawareness**, Equalized Odds gap 0.531 → 0.411, balanced accuracy 0.734 → 0.722.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.535 → **0.392** compared with accuracy-driven model selection, balanced accuracy 0.734 → 0.721.
• Statistical significance: 10 of 10 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).
• External validation (NHANES 2015–16): baseline balanced accuracy 0.740–0.749, Equalized Odds gap 0.453–0.508; best mitigated gap 0.021.

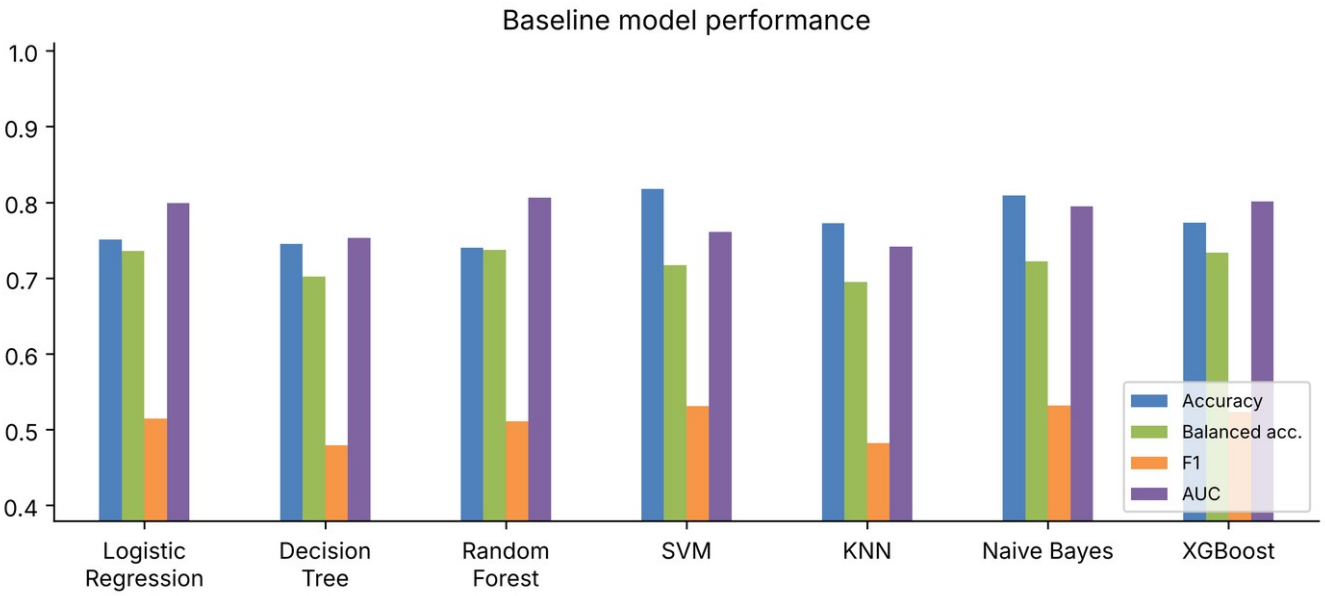
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
Elderly (>=60)	1827	36.1	626	34.3	0.130
Non-elderly (<60)	3236	63.9	313	9.7	0.160
All	5063	100	939	18.5	0.150


NHANES 2017-2018 (US CDC): Age group
Patients per group
Patients
3000
2500
2000
1500
1000
500
0
Elderly
(>=60)
Non-elderly
(<60)
CKD prevalence per group
%
35
30
25
20
15
10
5
0
34.3%
9.7%
Elderly
(>=60)
Non-elderly
(<60)

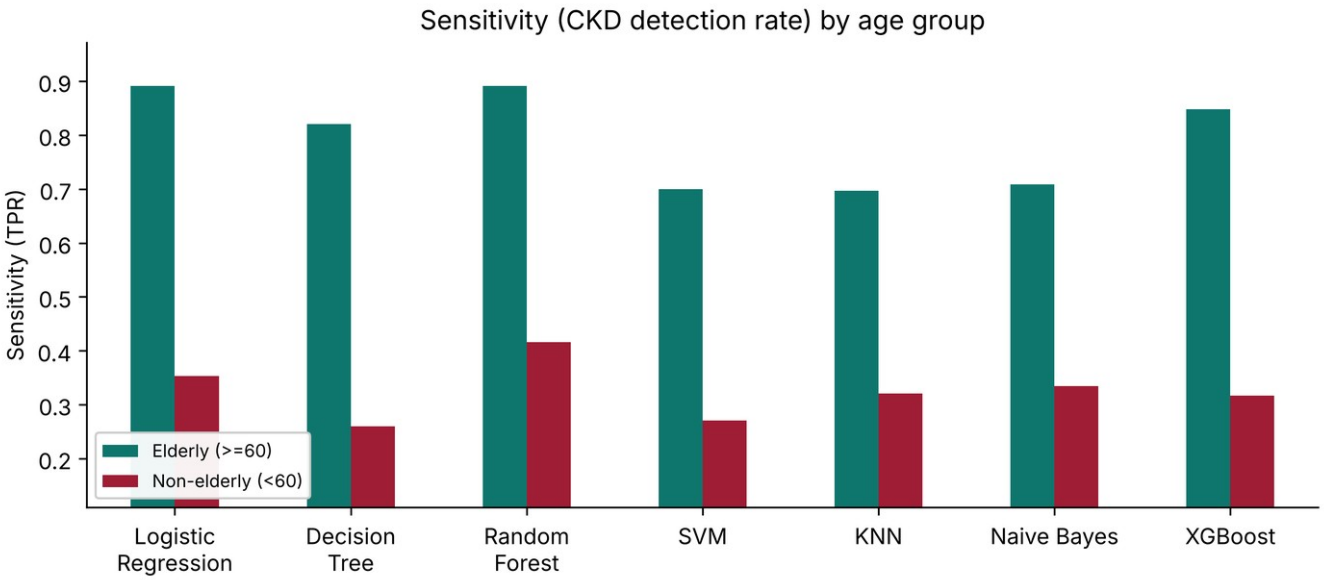
2. Baseline model performance

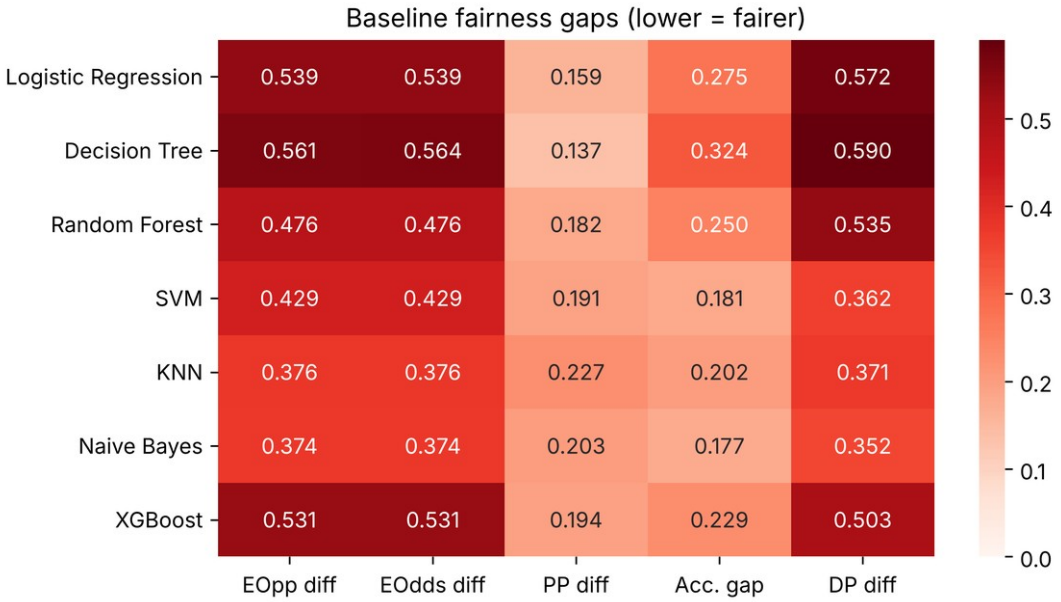
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.751 ± 0.002	0.403 ± 0.003	0.712 ± 0.003	0.515 ± 0.003	0.799 ± 0.001	0.736 ± 0.002
Decision Tree	0.745 ± 0.004	0.386 ± 0.004	0.633 ± 0.015	0.479 ± 0.005	0.753 ± 0.003	0.702 ± 0.005
Random Forest	0.740 ± 0.002	0.392 ± 0.003	0.733 ± 0.009	0.511 ± 0.004	0.806 ± 0.003	0.737 ± 0.004
SVM	0.818 ± 0.002	0.508 ± 0.006	0.557 ± 0.007	0.531 ± 0.006	0.761 ± 0.001	0.717 ± 0.004
KNN	0.773 ± 0.002	0.417 ± 0.004	0.571 ± 0.008	0.482 ± 0.005	0.741 ± 0.005	0.695 ± 0.004
Naive Bayes	0.809 ± 0.002	0.488 ± 0.004	0.584 ± 0.003	0.532 ± 0.003	0.794 ± 0.001	0.722 ± 0.002
XGBoost	0.773 ± 0.002	0.428 ± 0.004	0.671 ± 0.004	0.523 ± 0.003	0.801 ± 0.003	0.734 ± 0.002



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.572 ± 0.001	0.175 ± 0.001	0.539 ± 0.007	0.539 ± 0.007	0.159 ± 0.006	0.275 ± 0.001
Decision Tree	0.590 ± 0.033	0.135 ± 0.020	0.561 ± 0.028	0.564 ± 0.024	0.137 ± 0.035	0.324 ± 0.022
Random Forest	0.535 ± 0.005	0.223 ± 0.003	0.476 ± 0.012	0.476 ± 0.012	0.182 ± 0.005	0.250 ± 0.003
SVM	0.362 ± 0.004	0.167 ± 0.007	0.429 ± 0.009	0.429 ± 0.009	0.191 ± 0.008	0.181 ± 0.004
KNN	0.371 ± 0.010	0.244 ± 0.007	0.376 ± 0.011	0.376 ± 0.011	0.227 ± 0.015	0.202 ± 0.009
Naive Bayes	0.352 ± 0.002	0.213 ± 0.002	0.374 ± 0.006	0.374 ± 0.006	0.203 ± 0.005	0.177 ± 0.002
XGBoost	0.503 ± 0.005	0.178 ± 0.004	0.531 ± 0.012	0.531 ± 0.012	0.194 ± 0.009	0.229 ± 0.004
Perfect classifier (reference)	0.246	0.282	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by age group
Sensitivity (TPR)
0.9
0.8
0.7
0.6
0.5
0.4
0.3
0.2
Elderly (>=60)
Non-elderly (<60)
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.539 0.539 0.159 0.275 0.572
Decision Tree 0.561 0.564 0.137 0.324 0.590
Random Forest 0.476 0.476 0.182 0.250 0.535
SVM 0.429 0.429 0.191 0.181 0.362
KNN 0.376 0.376 0.227 0.202 0.371
Naive Bayes 0.374 0.374 0.203 0.177 0.352
XGBoost 0.531 0.531 0.194 0.229 0.503
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.5
0.4
0.3
0.2
0.1
0.0

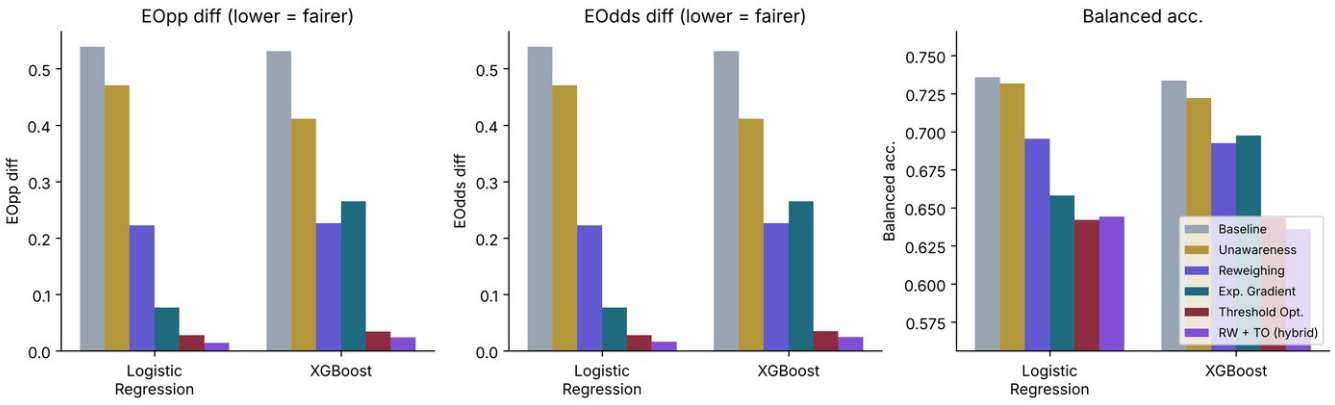
4. Bias mitigation

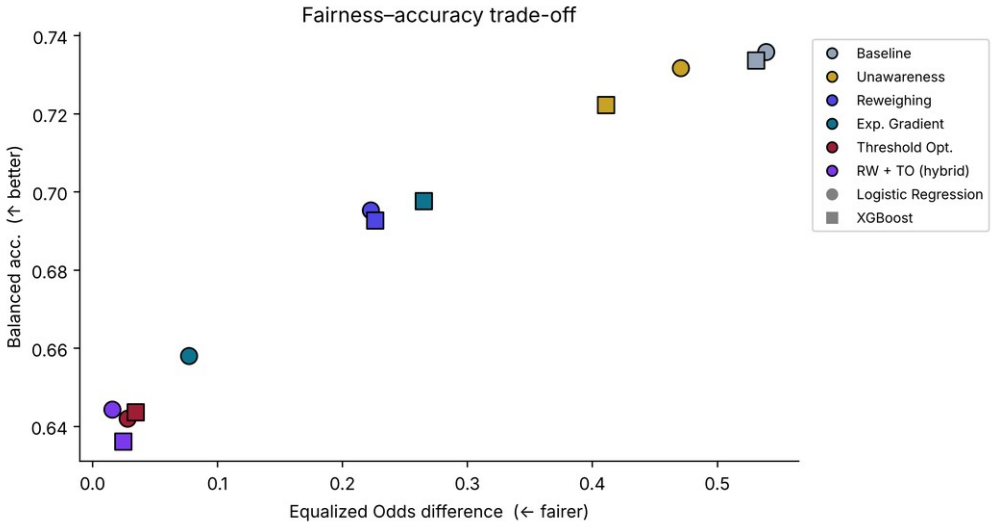
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.751 ± 0.002	0.736 ± 0.002	0.712 ± 0.003	0.515 ± 0.003
Logistic Regression	Unawareness	0.762 ± 0.001	0.732 ± 0.002	0.683 ± 0.002	0.516 ± 0.002
Logistic Regression	Reweighing	0.720 ± 0.002	0.695 ± 0.003	0.656 ± 0.005	0.465 ± 0.003
Logistic Regression	Exp. Gradient	0.691 ± 0.004	0.658 ± 0.003	0.605 ± 0.003	0.421 ± 0.003
Logistic Regression	Threshold Opt.	0.697 ± 0.022	0.642 ± 0.007	0.554 ± 0.023	0.405 ± 0.011
Logistic Regression	RW + TO (hybrid)	0.729 ± 0.013	0.644 ± 0.003	0.510 ± 0.018	0.411 ± 0.004
XGBoost	Baseline	0.773 ± 0.002	0.734 ± 0.002	0.671 ± 0.004	0.523 ± 0.003
XGBoost	Unawareness	0.770 ± 0.001	0.722 ± 0.003	0.647 ± 0.007	0.510 ± 0.003
XGBoost	Reweighing	0.747 ± 0.005	0.693 ± 0.007	0.606 ± 0.010	0.470 ± 0.009
XGBoost	Exp. Gradient	0.783 ± 0.003	0.698 ± 0.004	0.562 ± 0.011	0.490 ± 0.005
XGBoost	Threshold Opt.	0.697 ± 0.025	0.644 ± 0.005	0.559 ± 0.030	0.406 ± 0.007
XGBoost	RW + TO (hybrid)	0.684 ± 0.020	0.636 ± 0.005	0.560 ± 0.027	0.397 ± 0.006

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.539 ± 0.007	0.539 ± 0.007	0.159 ± 0.006	0.572 ± 0.001	0.175 ± 0.001	0.275 ± 0.001
Logistic Regression	Unawareness	0.471 ± 0.008	0.471 ± 0.008	0.215 ± 0.007	0.468 ± 0.004	0.226 ± 0.003	0.210 ± 0.004
Logistic Regression	Reweighing	0.223 ± 0.015	0.223 ± 0.015	0.308 ± 0.005	0.246 ± 0.006	0.502 ± 0.009	0.090 ± 0.003
Logistic Regression	Exp. Gradient	0.077 ± 0.016	0.077 ± 0.016	0.361 ± 0.011	0.093 ± 0.015	0.773 ± 0.034	0.014 ± 0.013
Logistic Regression	Threshold Opt.	0.027 ± 0.016	0.028 ± 0.016	0.345 ± 0.026	0.077 ± 0.008	0.793 ± 0.019	0.040 ± 0.009
Logistic Regression	RW + TO (hybrid)	0.014 ± 0.015	0.016 ± 0.014	0.351 ± 0.017	0.074 ± 0.008	0.769 ± 0.028	0.065 ± 0.016
XGBoost	Baseline	0.531 ± 0.012	0.531 ± 0.012	0.194 ± 0.009	0.503 ± 0.005	0.178 ± 0.004	0.229 ± 0.004
XGBoost	Unawareness	0.411 ± 0.014	0.411 ± 0.014	0.247 ± 0.009	0.395 ± 0.009	0.265 ± 0.005	0.176 ± 0.005
XGBoost	Reweighing	0.226 ± 0.025	0.226 ± 0.025	0.380 ± 0.007	0.179 ± 0.009	0.560 ± 0.016	0.046 ± 0.005
XGBoost	Exp. Gradient	0.265 ± 0.027	0.265 ± 0.027	0.366 ± 0.007	0.212 ± 0.010	0.436 ± 0.022	0.086 ± 0.007
XGBoost	Threshold Opt.	0.034 ± 0.029	0.035 ± 0.028	0.353 ± 0.023	0.074 ± 0.008	0.801 ± 0.021	0.033 ± 0.010
XGBoost	RW + TO (hybrid)	0.024 ± 0.017	0.025 ± 0.017	0.348 ± 0.016	0.059 ± 0.006	0.843 ± 0.016	0.023 ± 0.016





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Unawareness	-0.071	[-0.100, -0.046]	0.000	-0.003	[-0.011, 0.004]	0.388
Logistic Regression	Reweighing	-0.325	[-0.378, -0.275]	0.000	-0.039	[-0.052, -0.025]	0.000
Logistic Regression	Exp. Gradient	-0.466	[-0.522, -0.409]	0.000	-0.080	[-0.098, -0.062]	0.000
Logistic Regression	Threshold Opt.	-0.509	[-0.565, -0.459]	0.000	-0.101	[-0.119, -0.083]	0.000
Logistic Regression	RW + TO (hybrid)	-0.519	[-0.577, -0.452]	0.000	-0.091	[-0.110, -0.072]	0.000
XGBoost	Unawareness	-0.105	[-0.146, -0.065]	0.000	-0.009	[-0.020, 0.002]	0.122
XGBoost	Reweighing	-0.305	[-0.354, -0.258]	0.000	-0.039	[-0.052, -0.025]	0.000
XGBoost	Exp. Gradient	-0.287	[-0.330, -0.245]	0.000	-0.030	[-0.043, -0.016]	0.000
XGBoost	Threshold Opt.	-0.488	[-0.545, -0.412]	0.000	-0.093	[-0.111, -0.075]	0.000
XGBoost	RW + TO (hybrid)	-0.487	[-0.548, -0.406]	0.000	-0.099	[-0.117, -0.081]	0.000

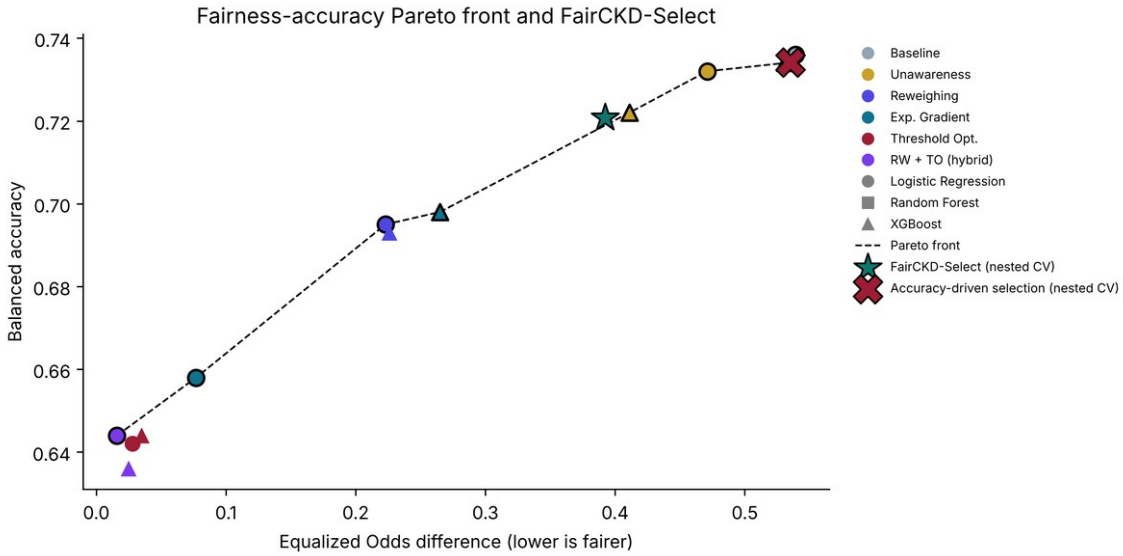
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.770 ± 0.003	0.721 ± 0.000	0.642 ± 0.005	0.392 ± 0.001	0.392 ± 0.001	0.378 ± 0.013
Accuracy-driven selection	0.756 ± 0.009	0.734 ± 0.002	0.699 ± 0.020	0.535 ± 0.009	0.535 ± 0.009	0.548 ± 0.035

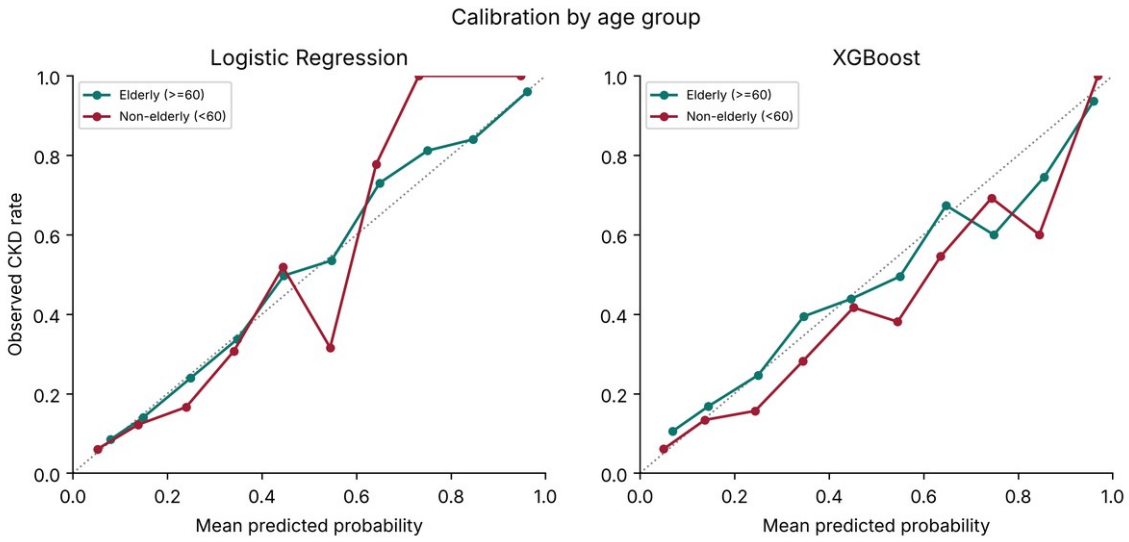
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
XGBoost	Exp. Gradient	1
XGBoost	Unawareness	9



4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
Elderly (>=60)	1827	34.264	33.797	-0.467	0.019	0.170	Logistic Regression
Non-elderly (<60)	3236	9.672	9.931	0.258	0.018	0.079	Logistic Regression
Elderly (>=60)	1827	34.264	33.982	-0.281	0.033	0.171	XGBoost
Non-elderly (<60)	3236	9.672	9.757	0.085	0.016	0.080	XGBoost

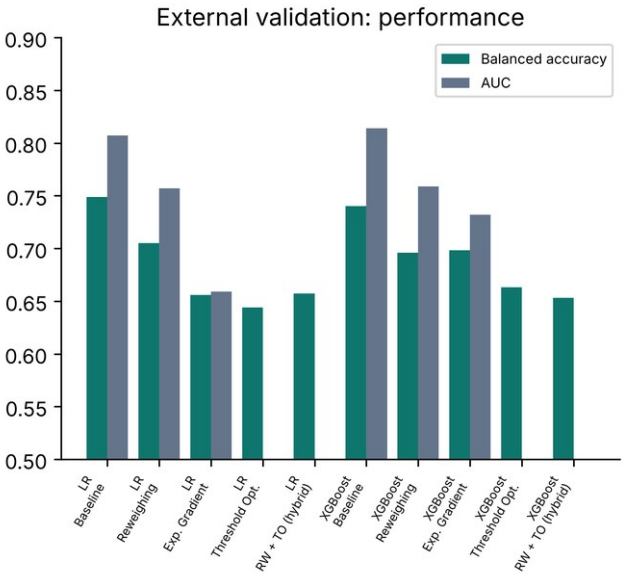


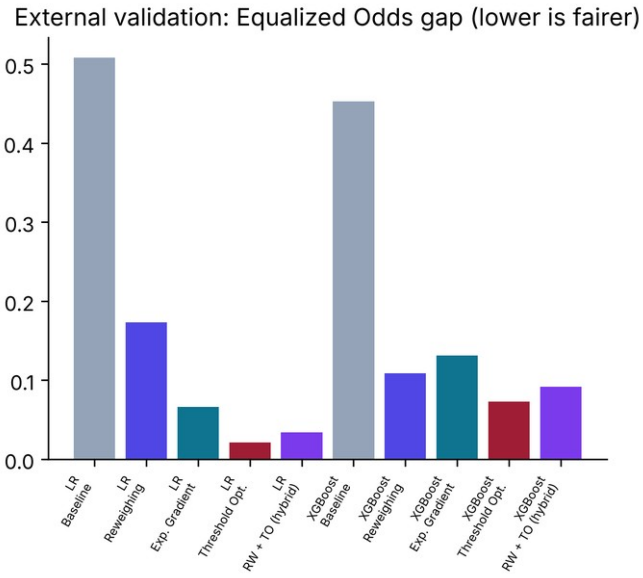
4d. Population-weighted estimates (NHANES survey weights)

Model	W Accuracy	W Balanced acc.	W EOpp diff	W EOdds diff	W DP diff	W TPR [Elderly (>=60)]	W TPR [Non-elderly (<60)]
Logistic Regression	0.787	0.731	0.584	0.584	0.567	0.907	0.323
Decision Tree	0.781	0.690	0.572	0.574	0.583	0.814	0.243
Random Forest	0.780	0.735	0.507	0.507	0.518	0.895	0.387
SVM	0.849	0.700	0.424	0.424	0.333	0.679	0.255
KNN	0.808	0.694	0.346	0.346	0.347	0.688	0.342
Naive Bayes	0.847	0.712	0.355	0.355	0.302	0.680	0.324
XGBoost	0.805	0.721	0.546	0.546	0.484	0.843	0.297

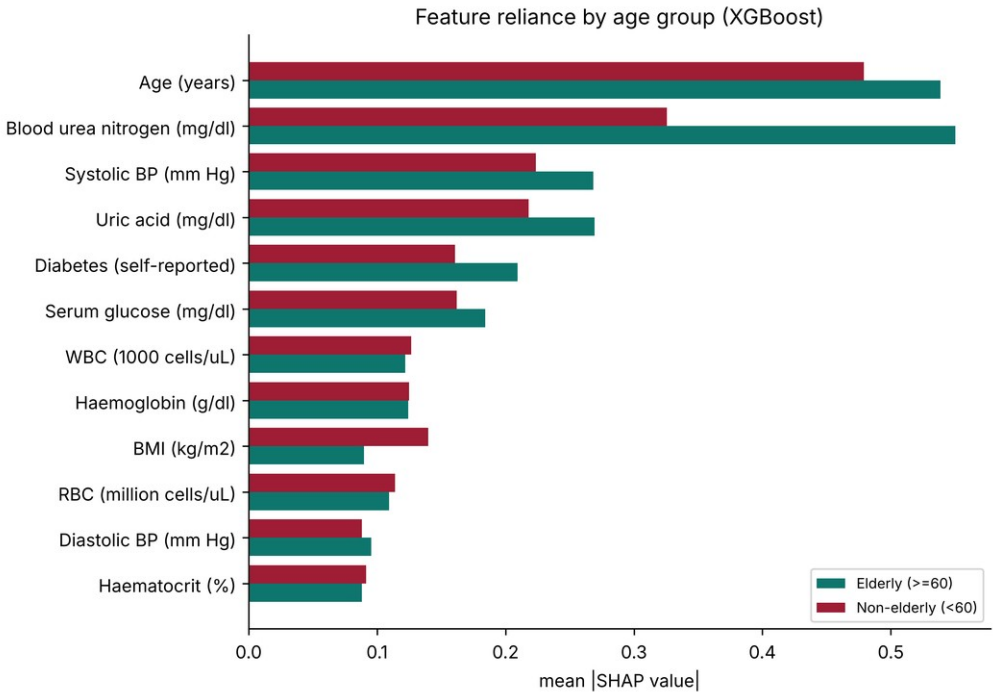
4e. External validation (trained on NHANES 2017–18, tested on NHANES 2015–16)

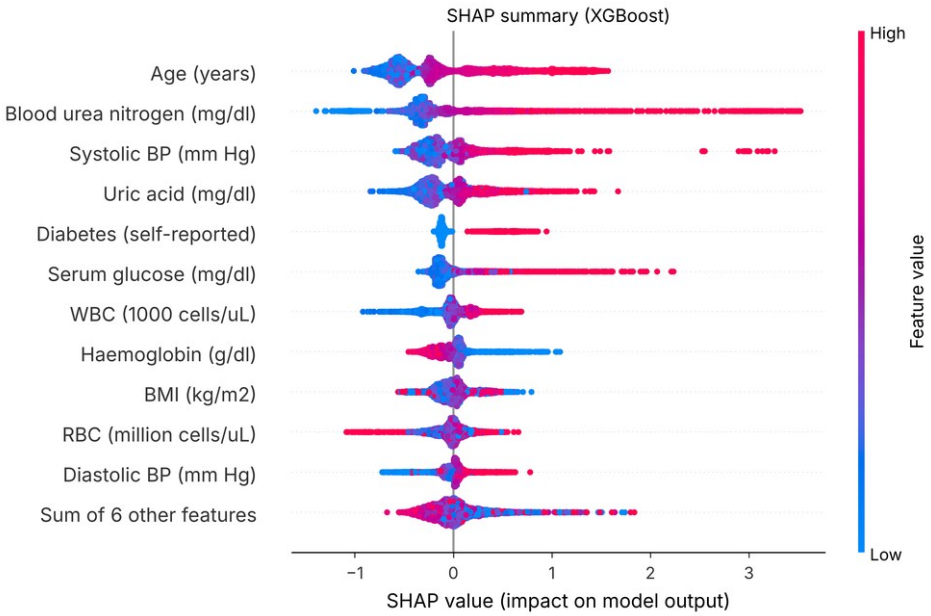
Model	Method	Accuracy	Balanced acc.	Recall	AUC	EOpp diff	EOdds diff
Logistic Regression	Baseline	0.784	0.749	0.696	0.807	0.475	0.508
Logistic Regression	Reweighing	0.742	0.705	0.650	0.757	0.173	0.173
Logistic Regression	Exp. Gradient	0.682	0.656	0.618	0.659	0.017	0.066
Logistic Regression	Threshold Opt.	0.673	0.644	0.601		0.009	0.021
Logistic Regression	RW + TO (hybrid)	0.712	0.657	0.576		0.034	0.034
XGBoost	Baseline	0.791	0.740	0.666	0.814	0.453	0.453
XGBoost	Reweighing	0.744	0.696	0.626	0.759	0.109	0.109
XGBoost	Exp. Gradient	0.791	0.698	0.561	0.732	0.131	0.131
XGBoost	Threshold Opt.	0.696	0.663	0.614		0.073	0.073
XGBoost	RW + TO (hybrid)	0.668	0.653	0.630		0.092	0.092





6. Explainability (SHAP)





Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.